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Print Blocking Won't Work - Permission to Print Part 2

ESPAÑOL

This is the second post in a series on 3D print blocking, for the first entry check out: [Print Blocking is Anti-Consumer – Permission to Print Part 1](#)

[Legislators across the U.S.](#) are [proposing laws](#) to force “print blockers” on 3D printers sold in their states. This mandated censorware is doomed to fail for its intended purpose, but will still manage to hurt the professional and hobbyist communities relying on these tools.

3D printers are commonly used to [repair belongings](#), [decorate homes](#), [print figurines](#), and so [much more](#). It’s not just hobbyists; 3D printers are also used professionally for parts prototyping and [fixturing](#), small-batch manufacturing, and [workspace organization](#). In rare cases, they’ve also been used to print parts needed for firearm assembly.

Many states have already banned manufacturing firearms using computer controlled machine tools, which are called “Computer Numerical Control or CNC machines,” and 3D printers without a license. Recently proposed laws seek to impose technical limitations onto 3D printers (and in some cases, CNC machines) in the hope of enforcing this prohibition.

This is a terrible idea; these mandates will be onerous to implement and will lock printer users into vendor software, impose one-time and ongoing costs on both printer vendors and users, and lay the foundation for a 3D-print censorship platform to be used in other jurisdictions. We dive more into these issues in [the first part of this series](#).

On a pragmatic level, however, these state mandates are just wishful thinking. Below, we dive into how 3D printing works, why these laws won't deter the printing of firearms, and how regular lawful use will be caught in the proposed dragnet.

How 3D Printers Work

To understand the impact of this proposed legislation, we need to know a bit about how 3D printers work. The most common printers work similarly to a computer-controlled hot glue gun on a motion platform; they follow basic commands to maintain temperature, extrude (push) plastic through a nozzle, and move a platform. These motions together build up layers to make a final "print." Modern 3D printers often offer more features like Wi-Fi connectivity or camera monitoring, but fundamentally they are very simple machines.

The basic instructions used by most 3D printers are called Geometric Code, or G-Code, which specify very basic motions such as "move from position A to position B while extruding plastic." The list of commands that will eventually print up a part are transferred to the printer in a text file thousands-to-millions of lines long. The printer dutifully follows these instructions with no overall idea of what it is printing.

While it is possible to write G-Code by hand for either a CNC machine or a 3D printer, the vast majority is generated by computer aided manufacturing (CAM) software, often called a "slicer" in 3D printing since it divides a 3D model into many 2D slices then generates motion instructions.

This same general process applies to CNC machines which use G-Code instructions to guide a metal removal tool. CNC machines have been included in previous prohibitions on firearm manufacturing and file distribution and are also targeted in some of these bills.

There are other types of 3D printers such as those that print [concrete](#), [resin](#), [metal](#), [chocolate](#) and other materials using slightly different methods. All of these would be subject to the proposed requirements regardless of how unlikely doing harm with a gun made out of chocolate would be.



Simple rectangular 3D model for test fit

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AirgradientTestPlate_0.4n_0.15mm_PETG_XLIS_1h54m.gcode x
10024 M204 P2500
10025 G1 E-.16 F2100
10026 ;WIPE_START
10027 G1 F19200
10028 G1 X210 Y277.887 E-.00613
10029 G1 X209.386 Y277.813 E-.006426
10030 G1 X208.787 Y277.572 E-.006709
10031 G1 X208.27 Y277.185 E-.00671
10032 G1 X207.872 Y276.677 E-.006706
10033 G1 X207.618 Y276.083 E-.006713
10034 G1 X207.527 Y275.444 E-.006707
10035 G1 X207.604 Y274.803 E-.006708
10036 G1 X207.836 Y274.224 E-.006481
10037 G1 X208.231 Y273.689 E-.00691
10038 G1 X208.482 Y273.492 E-.003317
10039 ;WIPE_END
10040 G1 X180.855 Y318.606 Z1.573 F24000
10041 G1 Z.65 F720
10042 G1 E.8 F1500
10043 ;TYPE:Perimeter
10044 G1 F2700
10045 G1 X180.723 Y318.641 E.00356
10046 G1 X180.136 Y318.718 E.01543
10047 G3 X179.277 Y318.641 I-.033 J-4.448 E.02251
10048 G1 X178.467 Y318.344 E.02248
10049 G1 X177.762 Y317.848 E.02246
10050 G3 X177.209 Y317.186 I3.122 J-3.169 E.02251
10051 G3 X176.846 Y316.403 I3.836 J-2.253 E.02252
10052 G3 X176.698 Y315.553 I4.288 J-1.184 E.02252
10053 G1 X176.698 Y315.28 E.00711
10054 G3 X176.846 Y314.43 I4.436 J.334 E.02252
10055 G1 X177.072 Y313.884 E.0154
10056 G3 X177.762 Y312.986 I3.328 J1.839 E.02962
10057 G3 X178.467 Y312.489 I2.9 J3.372 E.02251
10058 G3 X179.277 Y312.192 I1.929 J4.009 E.02251
10059 G1 X179.845 Y312.117 E.01493
10060 G1 X180.713 Y312.191 E.0227
10061 G3 X181.533 Y312.489 I-1.126 J4.372 E.02277
10062 G3 X182.238 Y312.986 I-2.438 J4.214 E.0225
10063 G3 X182.791 Y313.648 I-3.291 J3.31 E.02251
10064 G3 X183.154 Y314.43 I-3.836 J2.253 E.0225
10065 G3 X183.302 Y315.28 I-4.288 J1.184 E.02252
10066 G1 X183.302 Y315.553 E.00711
10067 G3 X183.154 Y316.403 I-4.436 J-.334 E.02252
10068 G1 X182.928 Y316.95 E.01542
10069 G3 X182.238 Y317.848 I-3.328 J-1.84 E.02962
10070 G1 X181.769 Y318.208 E.01541
10071 G1 X180.986 Y318.571 E.02249
10072 G1 X180.913 Y318.59 E.00197
10073 G1 X180.745 Y318.198 F24000
10074 G1 F2700

```

Part of a 173490 line long G-Code file produced by slicer for simple rectangular model.

Part of a 173,490 line long G-Code file for a simple rectangular part.

How is Firearm Detection Supposed to Work?

Under these proposed laws, manufacturers of consumer 3D printers must ensure their printers only work with their software, and implement firearm detection algorithms on either the printer itself or in a slicer software. These algorithms must detect firearm files using a maintained database of existing models. Vendors of printers must then verify that printers are on the allow-list maintained by the state before they can offer them for sale.

Owners of printers will be guilty of a crime if they circumvent these intrusive scanning procedures or load alternative software, which they might do because their printer manufacturer ends support. Owners of existing noncompliant 3D

printers in regulated states will be unable to resell their printers on the secondary market legally.

What Will Actually Happen?

While the proposed laws allow for scanning to happen on either the printer itself or in the slicer software, the reality is more complicated.

The computers inside many 3D printers have very limited computational and storage ability; it will be impossible for the printer's computer to render the G-Code into a 3D model to compare with the database of prohibited files. Thus the only way to achieve this through the machine would be to upload all printer files to a cloud comparison tool, creating new delays, errors, and unacceptable invasions of privacy.

Many vendors will instead choose to permanently link their printers to a specific slicer that implements firearm detection. This requires cryptographic signing of G-Code to ensure only authorized prints are completed, and will lock 3D printer owners into the slicer chosen by their printer vendor.

Regardless of the specifics of their implementation, these algorithms will interfere with 3D printers' ability to print other parts without actually stopping manufacture of guns. It takes very little skill for a user to make slight design tweaks to either a model or G-Code to evade detection. One can also design incomplete or heavily adorned models which can be made functional with some post-print alterations. While this would be pioneered by skilled users—like the ones who designed today's 3D printed guns—once the design and instructions are out there anyone able to print a gun today will be able to follow suit.

Firearm part identification features also impose costs onto 3D printer manufacturers, and hence their end consumers. 3D printer manufacturers must develop or license these costly algorithms and continuously maintain and update both the algorithm and the database of firearm models. Older printers that cannot comply will not be able to be resold in states where they are banned, creating additional E-waste.

While those wishing to create guns will still be able to do so, people printing other functional parts will likely be caught up in these algorithms, particularly for things like film props, kids' toys, or decorative models, which often closely resemble real firearms or firearm components.

What Are The Impacts of These Changes?

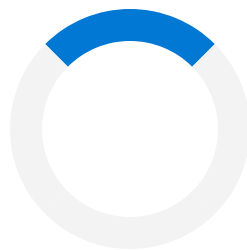
Technological restrictions on manufacturing tools' abilities are harmful for many reasons. EFF is particularly concerned with this regulation locking a 3D printer to proprietary vendor software. Vendors will be able to use this mandate to support only in-house materials, locking users into future purchases. Vendor slicer software is often based on out-of-date, open source software, and forcing users to use that software deprives them of new features or even use of their printer altogether if the vendor goes out of business. At worst, some of these bill will make it a misdemeanor to fix those problems and gain full control of your printer.

File-scanning frameworks required by this regulation will lay the foundation for future privacy and freedom intrusions. This requirement could be co-opted to scan prints for copyright violations and be abused similar to [DMCA takedowns](#), or to suppress models considered obscene by a patchwork of definitions. What if you were unable to print a repair part because the vendor asserted the model was in violation of their [trademark](#)? What if your print was considered obscene?

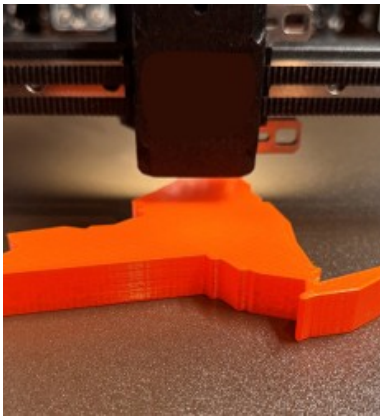
Regardless of your position on current prohibitions on firearms, we should all fight back against this effort to force technological restrictions on 3D printers, and legislators must similarly abandon the idea. These laws impose real costs and potential harms among lawful users, [lay the groundwork for future censorship](#), and simply won't deter firearm printing.

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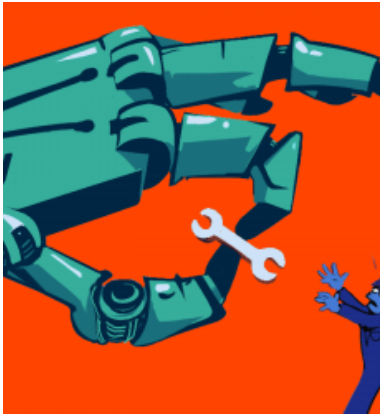
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